Complete Guide

Vector DB Explained

Learn RAG Vector Databases from Scratch.

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Let's build a Vector Database that understands meaning, not just words.

Here is the step-by-step Python guide.

The Stack

We will use:

1. **Python** (Logic)
2. **Sentence-Transformers** (Text to Vectors)
3. **ChromaDB** (The Vector Database)

```
pip install sentence-transformers chromadb
```


STEP 01

Turning Text into Numbers

We need a model to translate English into 'machine math' (vectors).

We will load all-MiniLM-L6-v2 from Hugging Face:

```
from sentence_transformers import SentenceTransformer
model = SentenceTransformer('all-MiniLM-L6-v2')
```


STEP 02

Setting Up ChromaDB

Next, we need a place to store these vectors and search them fast. We need to initialize a local client and create a Collection (think of this like a SQL table):

```
import chromadb
client = chromadb.Client()
collection = client.create_collection(name="my_kb")
```


STEP 03

Adding Documents

Let's feed the database some raw text about AI and Tech. This script converts the text to vectors and stores them with an ID:

```
docs = [
    "Artificial intelligence enables machines to learn and make decisions.",
    "Machine learning focuses on data."
]
embeddings = model.encode(docs)
collection.add(
    documents=docs,
    embeddings=embeddings,
    ids=["id1", "id2"]
)
```

STEP 04

Querying by Meaning

Now, we can ask a question that doesn't share keywords with our data:

```
 query = "What is artificial intelligence?"
query_vector = model.encode(query)

results = collection.query(
    query_embeddings=[query_vector],
    n_results=1
)
```


Semantic Search

Even though we didn't use the exact words, the DB will return:

> *"Artificial intelligence enables machines to learn and make decisions."*

The query "What is artificial intelligence?" semantically matches with our stored documents about AI.

Why Vector DBs Matter

1. Semantic Understanding: Find content by meaning, not just keywords
2. AI-Powered Search: Enables RAG, chatbots, and recommendation systems
3. Scalability: Handle millions of vectors with fast similarity search
4. Real-World Impact: Powers ChatGPT, Google Search, and modern AI apps

Thank You!

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